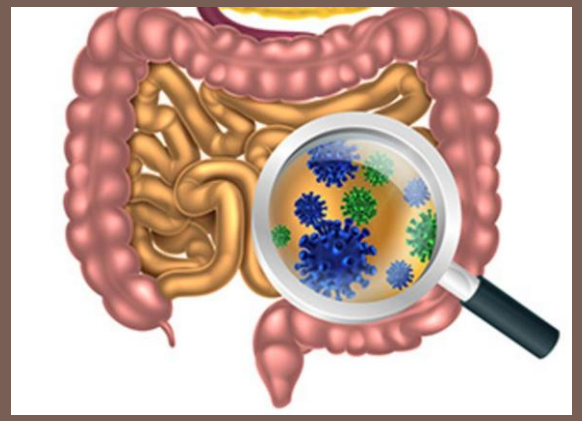




INTESTINAL MICROBIOLOGY

Assoc.Prof.Kanitha Patarakul, MD, PhD

Assist.Prof.Ekasit Kowitdamrong, MD

Case 1

In January, a 2-month-old formula-fed infant has fever, repeated vomiting, and profuse diarrhea for 2 days.

The mother declares he has no known medical problems.

Case 1

Physical examination reveals BT 39˚C, BP 80/50 mmHg, PR 160/min, RR 45/min





Which virus is the most likely causative agent?

A. Astrovirus

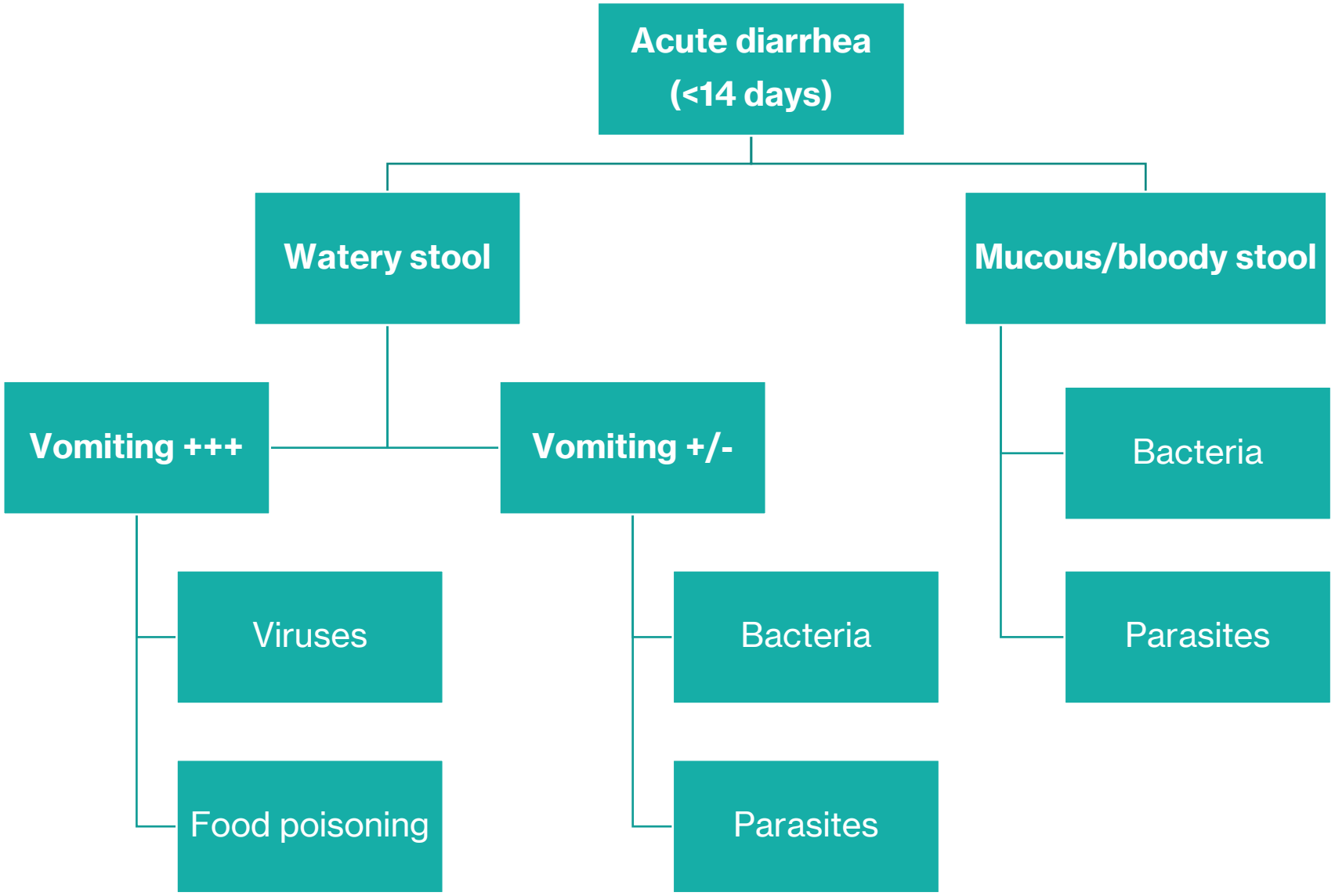
B. Norovirus

C. Rotavirus

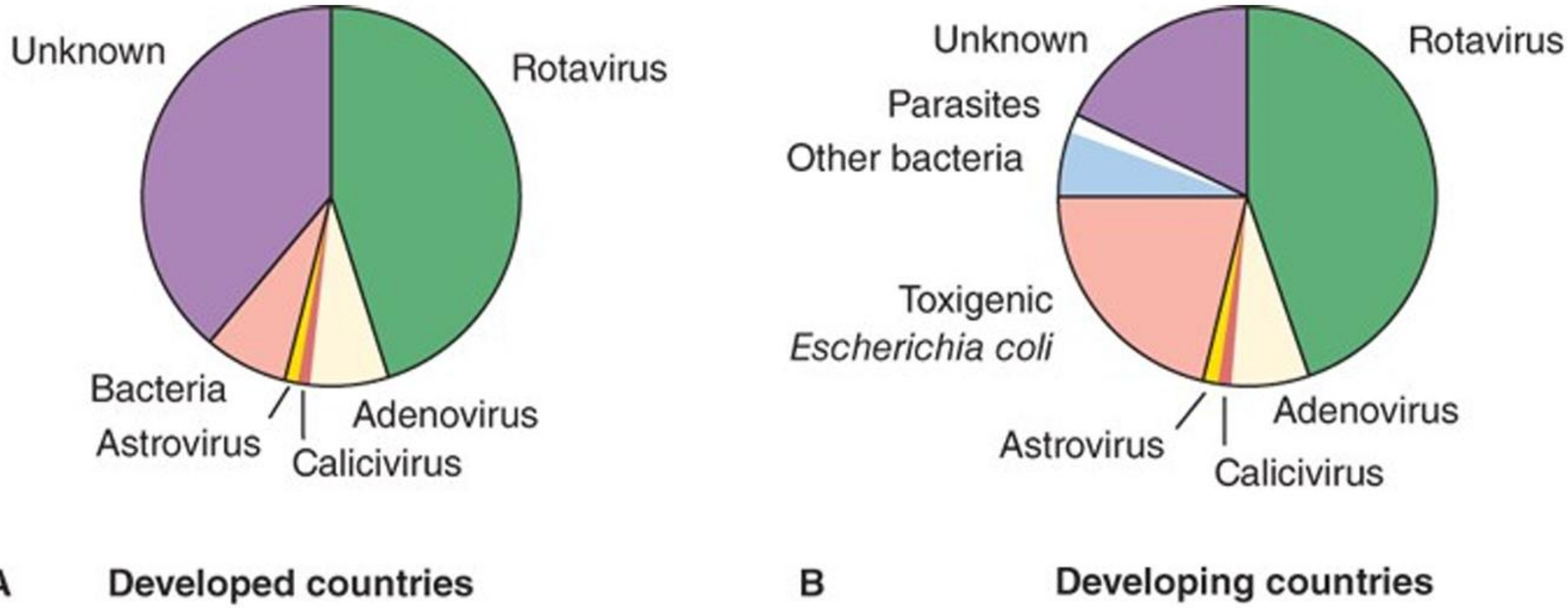
D. Sapovirus

E. Enteric adenovirus

Approach to acute diarrhea



Cause of gastroenteritis in hospitalized children



A **Developed countries** **B** **Developing countries**

Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e

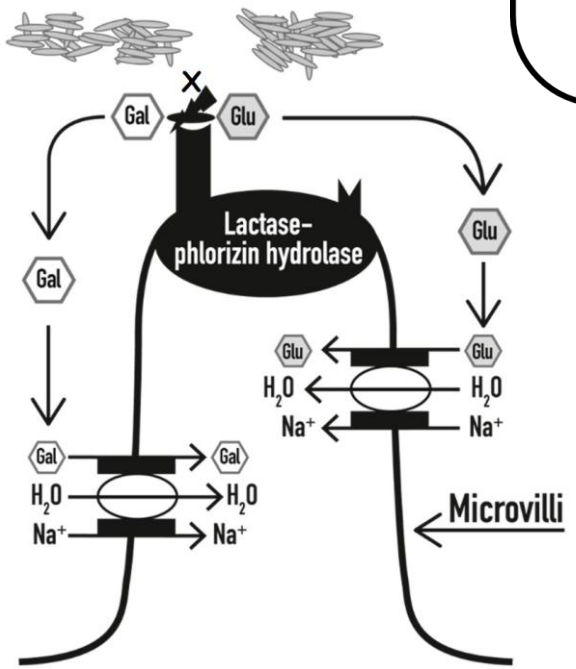
Copyright © McGraw-Hill Education. All rights reserved.

Differential diagnosis

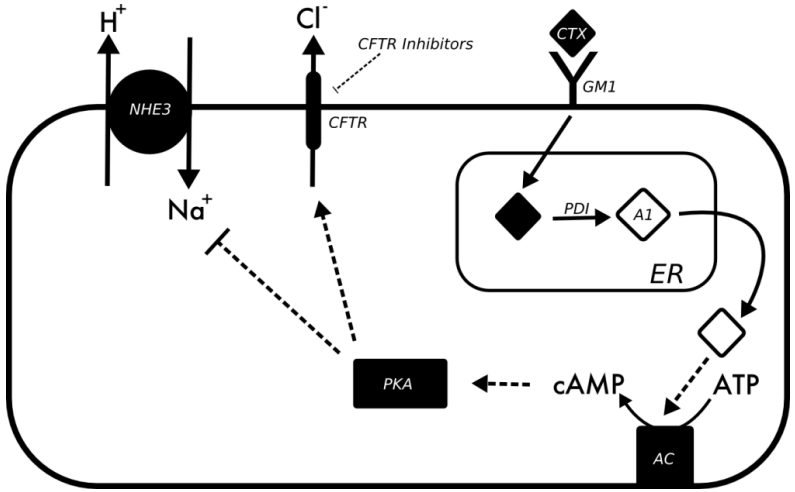
Factor	Differential clue	Likely viruses	Strength of clue
Season	Winter	R, N	++++
		As	+
Age	6-24 months	R, As, S	++
	<6 months	Ad	+
Duration	>7 days	Ad	+

Which of the following is the most likely pathogenesis?

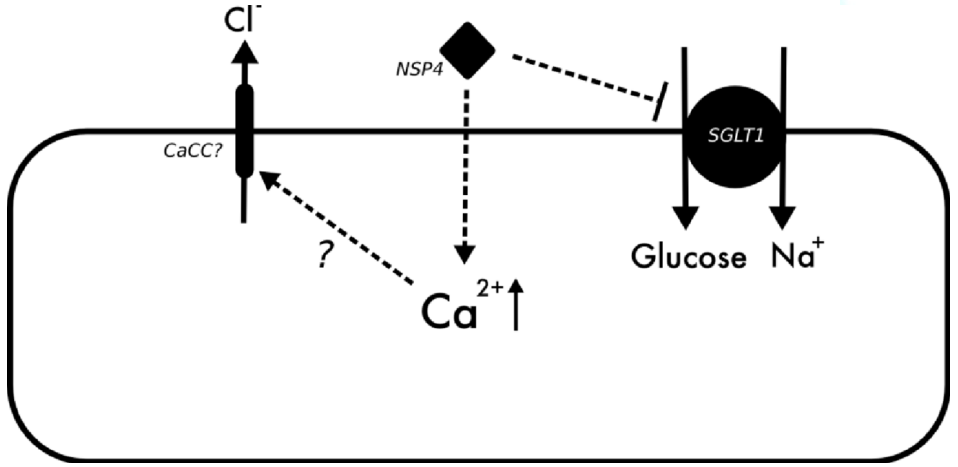
A.



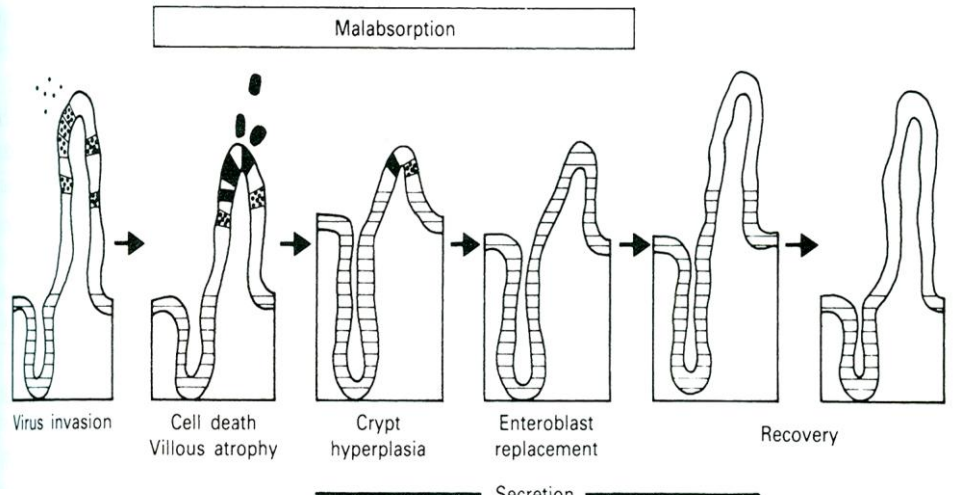
B.



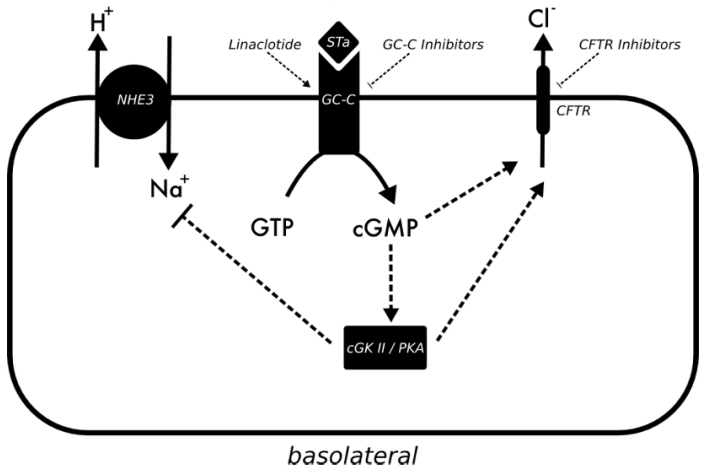
C.

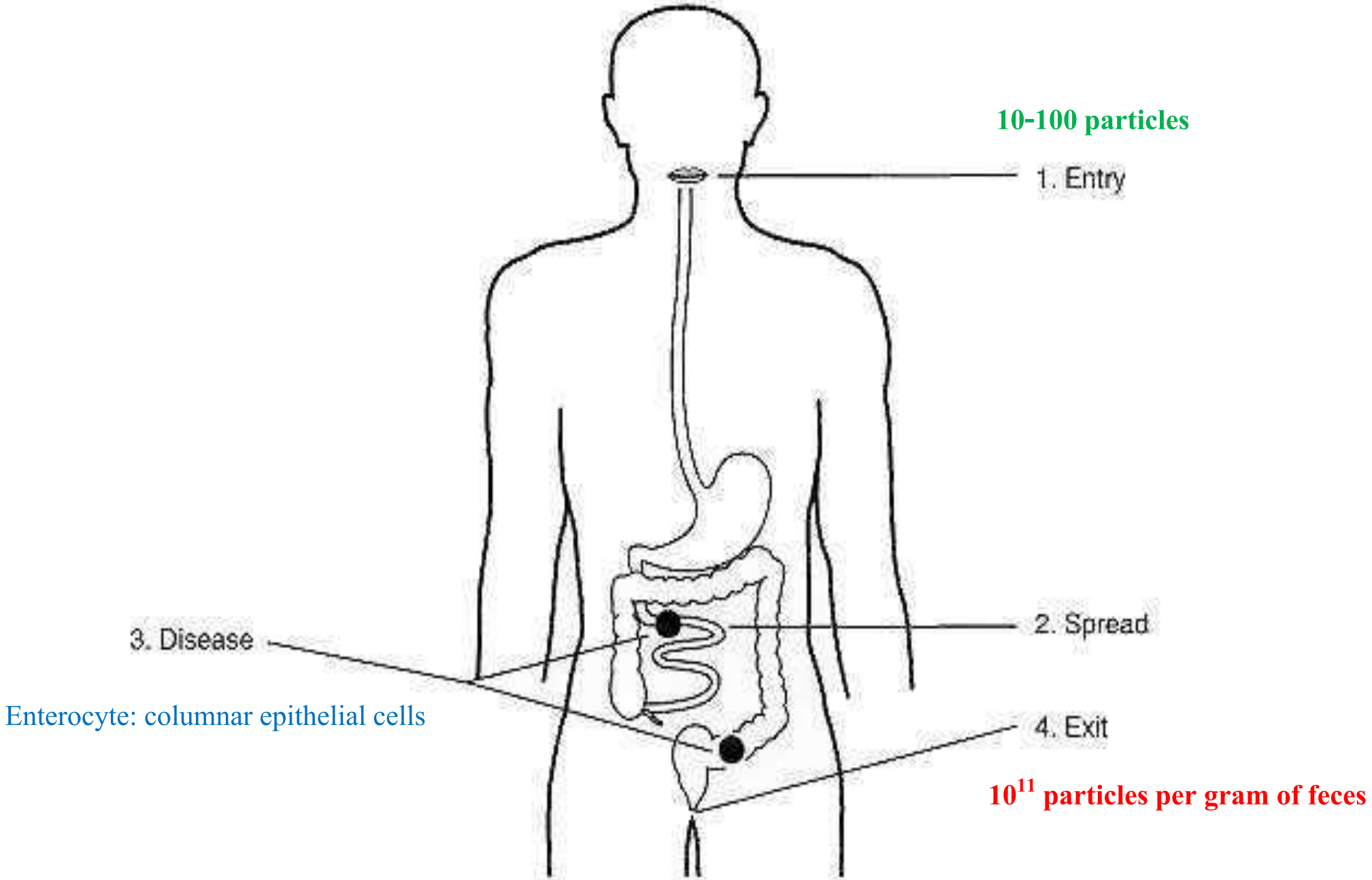


D.

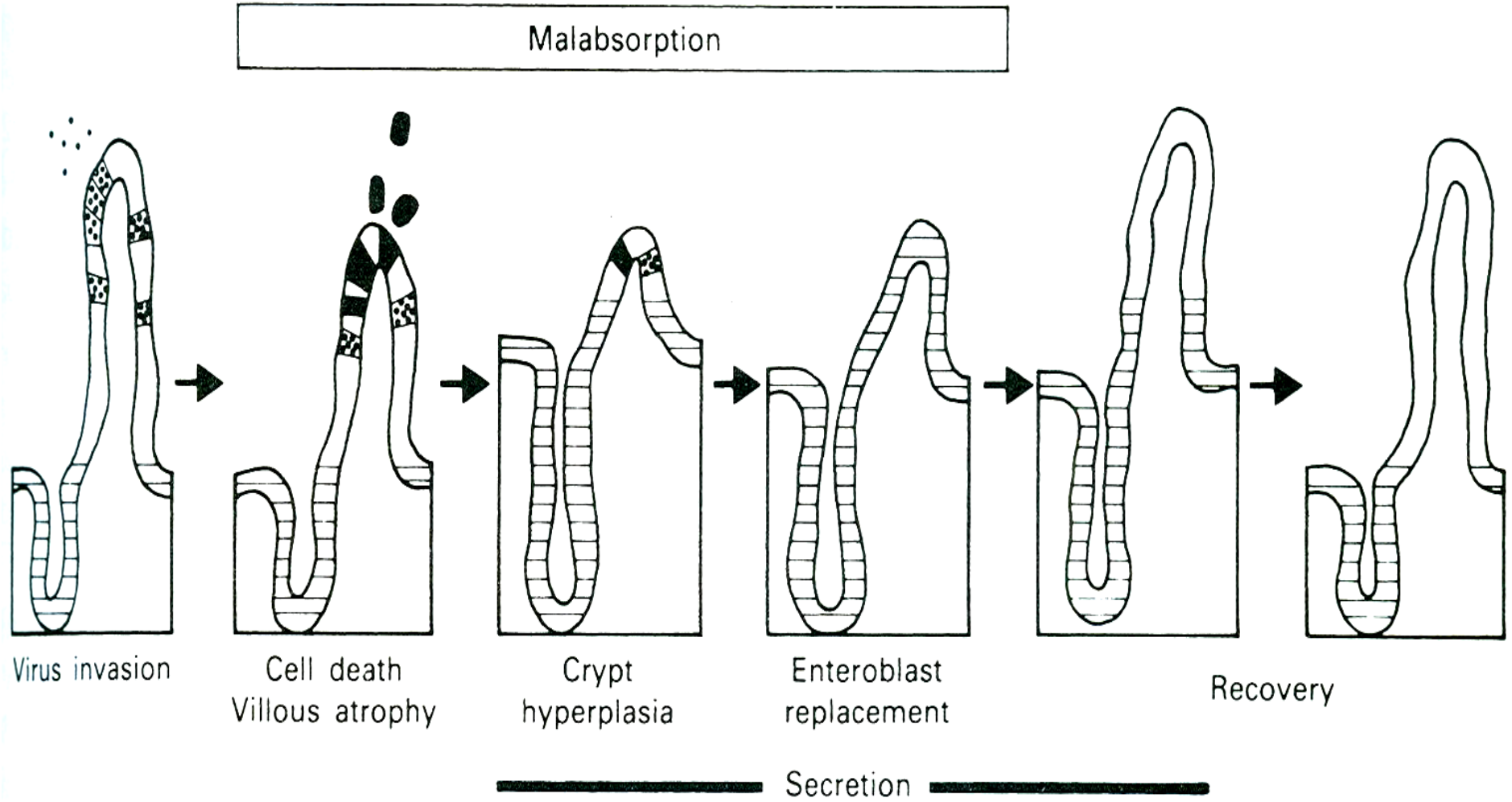


E.

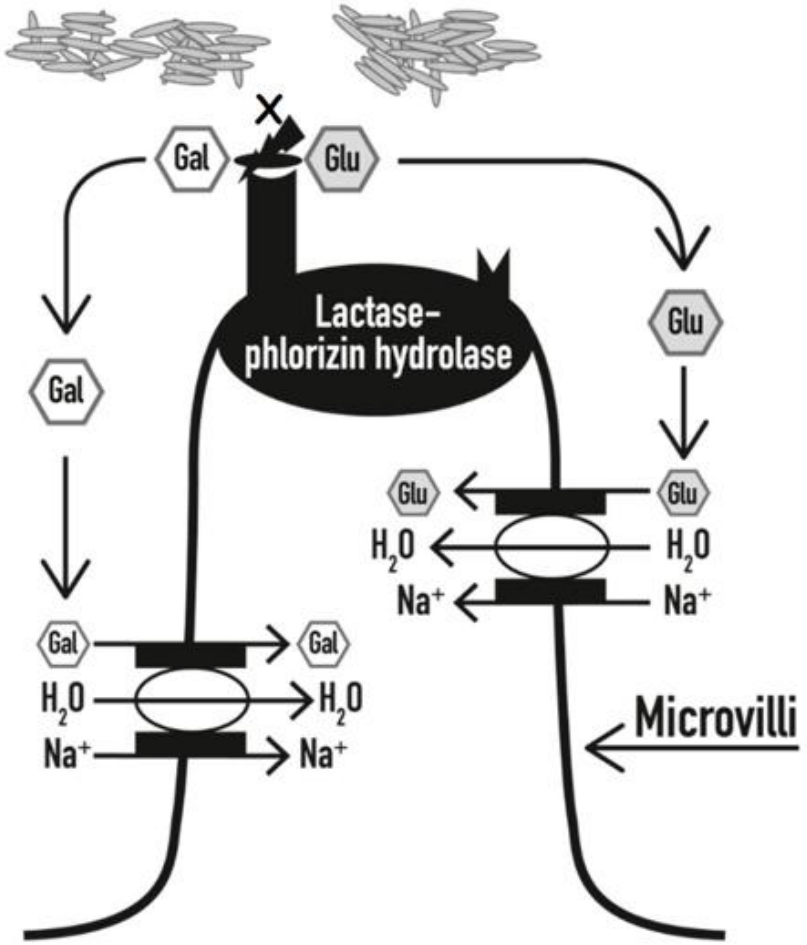



10-100 particles
1. Entry
2. Spread
3. Disease
4. Exit
Enterocyte: columnar epithelial cells
10^{11} particles per gram of feces

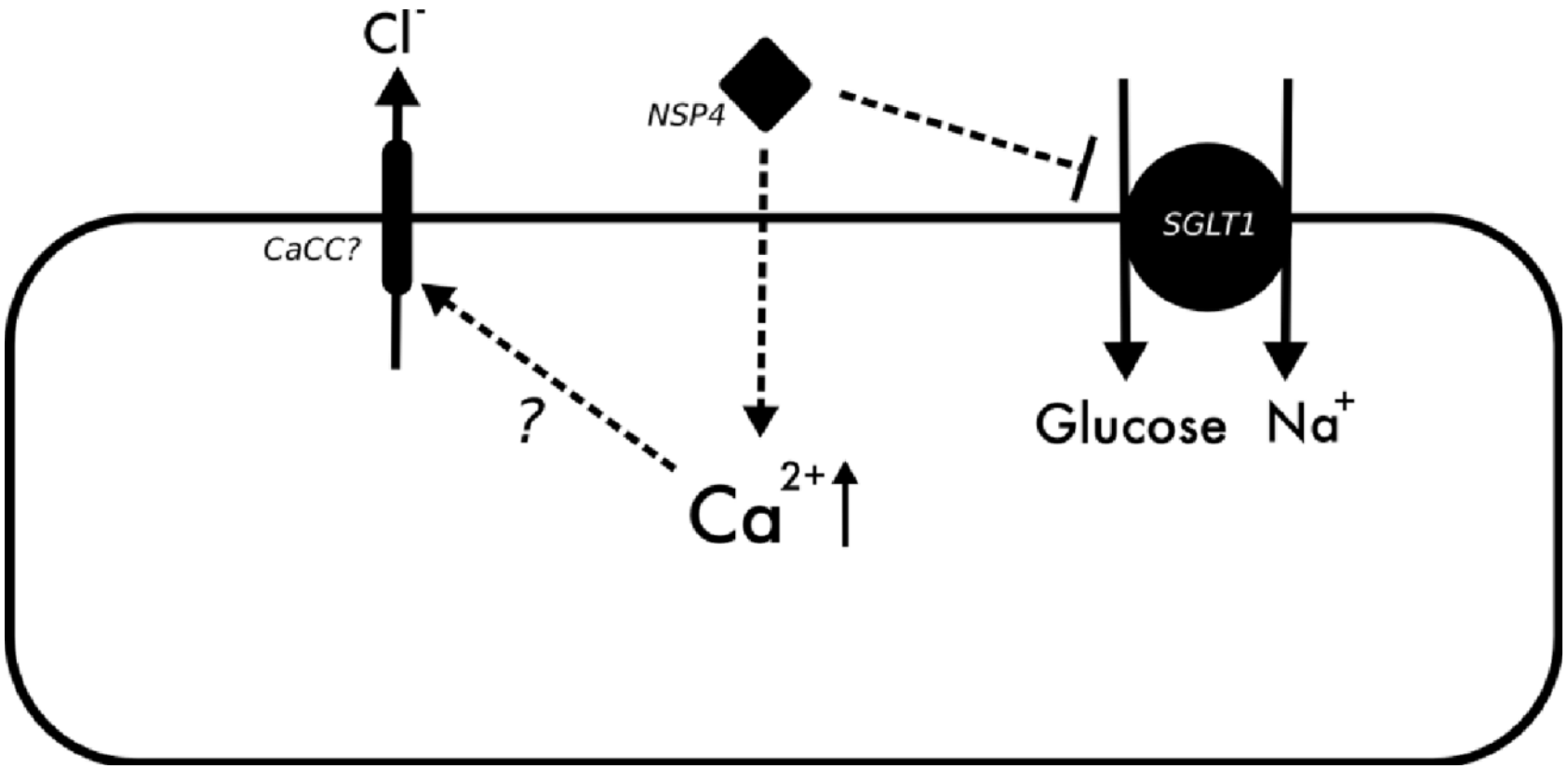
Rotavirus gastroenteritis: pathogenesis



Rotavirus gastroenteritis: pathogenesis



Rotavirus gastroenteritis: pathogenesis



Which of the following supports the diagnosis of this viral gastroenteritis?

A. Stool WBC 10 cells/HPF

B. Stool RBC 5 cells/HPF

C. Mucous stool

D. Stool pH 5.0

E. Negative stool reducing substance

Rotavirus: laboratory diagnosis

- Stool examination: no WBC, no RBC
- Stool pH: <5.5
- Stool reducing substance: ≥2+

What is the most appropriate laboratory investigation?

A. Electron microscopy

B. Viral culture

C. Antigen detection

D. Virus-specific IgM

E. RT-PCR

Rotavirus: laboratory diagnosis

- EM, Immune EM (IEM)
  - Samples: stool
- Viral antigen detection
  - Samples: stool
  - Methods: ELISA, LA, IC
- Viral isolation
  - Samples: stool
  - Methods: cell culture
- Antibody detection
  - Samples: serum
  - Methods: ELISA, NT
- Viral genome detection
  - Samples: stool
  - Methods: Gel electrophoresis, RT-PCR

What is the most effective preventive measure against this viral gastroenteritis?

A. Inactivated vaccine

B. Live-attenuated vaccine

C. Breastfeeding

D. Hand washing

E. Wearing mask

Rotavirus: prevention

- Rotavirus vaccine
- Hygiene and sanitation
- Breastfeeding campaign



Rotavirus vaccines

- Rotarix (GSK)
  - Oral live-attenuated vaccine
  - Human G1P[8]
  - 2 doses: at 2 and 4 months old
- RotaTeq (Merck)
  - Oral live-attenuated vaccine
  - Bovine Pentavalent G1-G4, P[8]
  - 3 doses: at 2, 4, and 6 months old
- Rotavac (Bharat Biologicals)
  - Oral live-attenuated vaccine
  - Human G9P[11]
  - 3 doses: at 2, 4, and 6 months old
- Rotasiil (Serum Institute of India)
  - Oral live-attenuated vaccine
  - Bovine Pentavalent G1-G4, G9
  - 3 doses: at 2, 4, and 6 months old

Case 2

In December, a 25-year-old female develops fever, frequent vomiting, and diarrhea for 2 days.

She declares no known medical problems.

Two days ago, she ate seafood salad with her friends at a party. Ten of her friends also suffered from similar illness.

Case 2

Physical examination reveals BT 38˚C, BP 100/60 mmHg, PR 80/min, RR 20/min





Which virus is the most likely causative agent?

A. Astrovirus

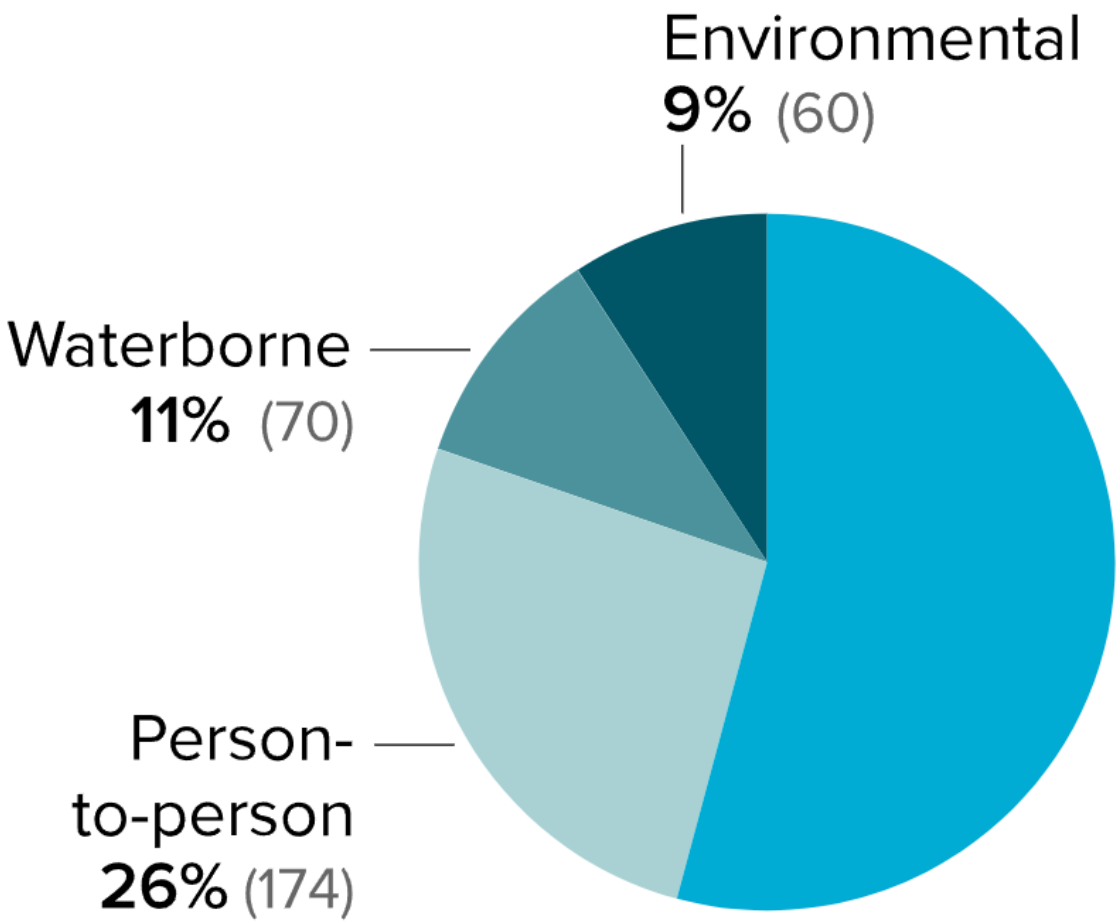
B. Norovirus

C. Rotavirus

D. Sapovirus

E. Enteric adenovirus

Norovirus: transmission



Epidemic

Shellfish
Salad

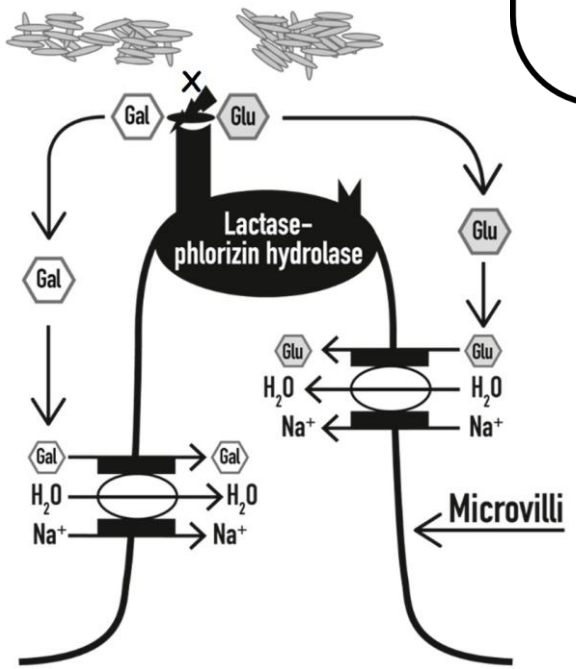
Peak in winter

Source: JE Matthews, et al. Epidemiology and infection, 2012.

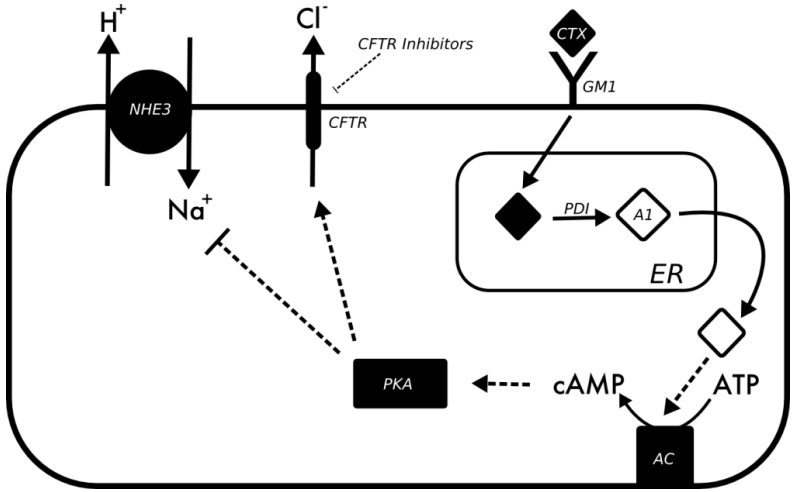
https://knowablemagazine.org/article/health-disease/2017/norovirus-perfect-pathogen

Which of the following is the most likely pathogenesis?

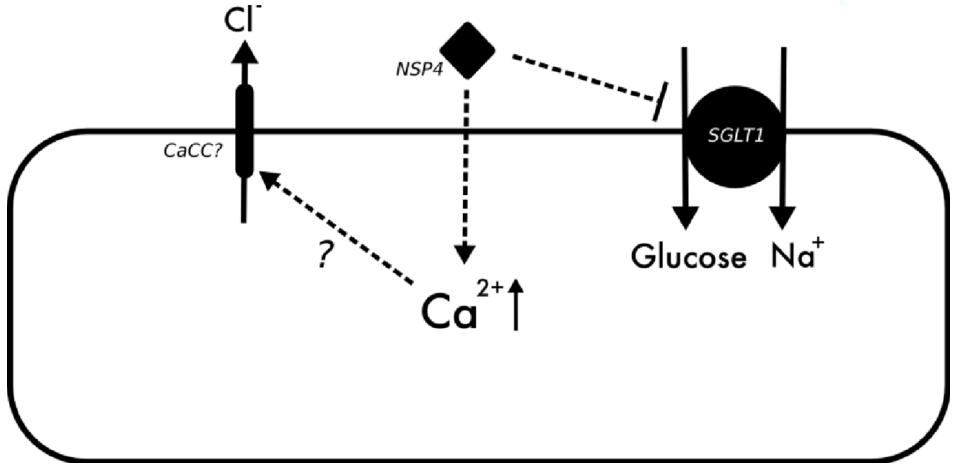
A.



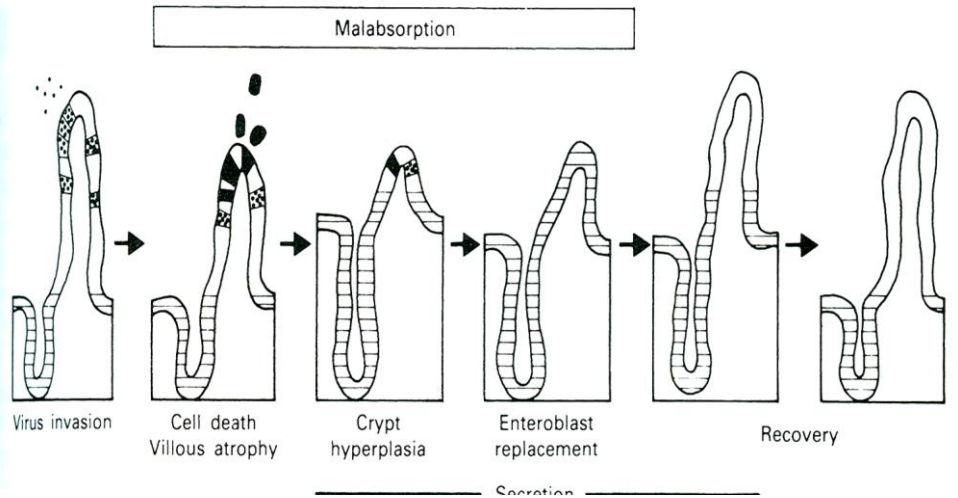
B.



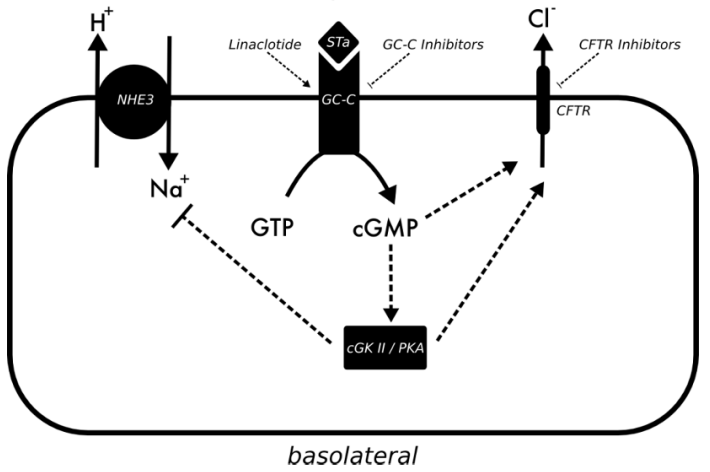
C.



D.



E.

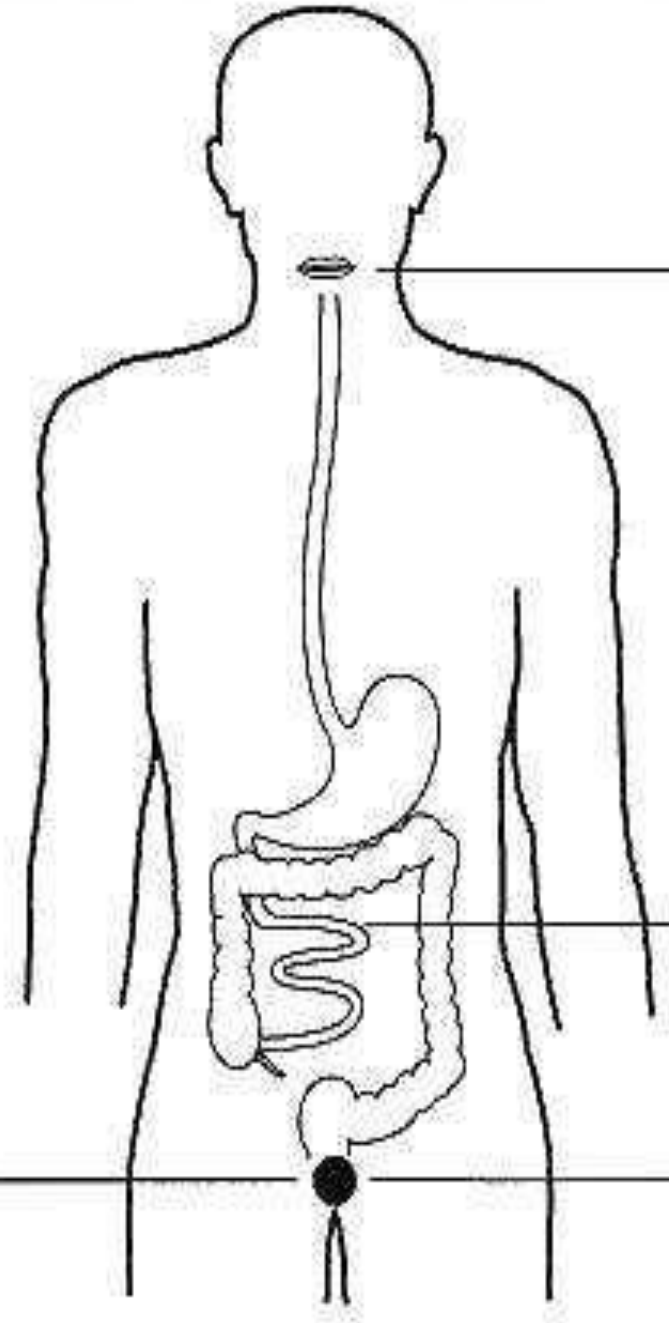


Blunting of intestinal villi

Crypt hyperplasia

↓ Small-intestinal brush border enzymes

Malabsorption



10-100 particles

1. Entry

2. Infection (small intestine)

Enterocyte: columnar epithelial cells

3. Disease Diarrhea

4. Exit

7×10^7 particles per gram of feces

What is the most appropriate laboratory investigation?

A. Electron microscopy

B. Viral culture

C. Antigen detection

D. Virus-specific IgM

E. RT-PCR

Norovirus: laboratory diagnosis

- EM, Immune EM (IEM)
  - Samples: stool
- Viral antigen detection
  - Samples: stool
  - Methods: ELISA, IC
- Viral isolation
  - Samples: stool
  - Methods: cell culture
- Antibody detection
  - Samples: serum
  - Methods: ELISA, NT
- Viral genome detection
  - Samples: stool
  - Methods: RT-PCR

Laboratory diagnosis

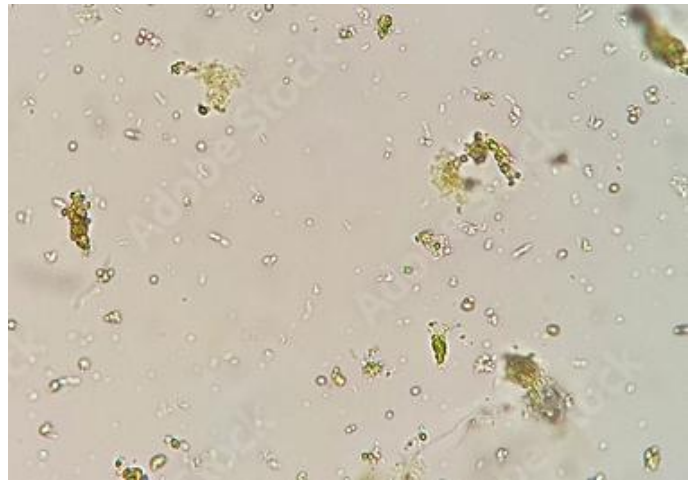
Viruses	Preferred methods
Rotaviruses	EM, Ag, RT-PCR
Astroviruses	EM, Ag, RT-PCR
Noroviruses	EM, Ag, RT-PCR
Sapoviruses	EM, Ag, RT-PCR
Enteric Adenoviruses	EM, Ag, PCR

Case 3



- A 10-year-old boy presents with a 2-day history of vomiting and profuse, rice-water diarrhea without abdominal pain.
- BT 36°C, BP 60/40 mmHg, HR 120/min, RR 24/min
- He is lethargic and has sunken eyes, a dry mouth, cold and clammy skin, and poor skin turgor.





Which of the following is the most likely causative agent?

A. Enteropathogenic *E. coli*

B. *Campylobacter jejuni*

C. *Salmonella* Typhi

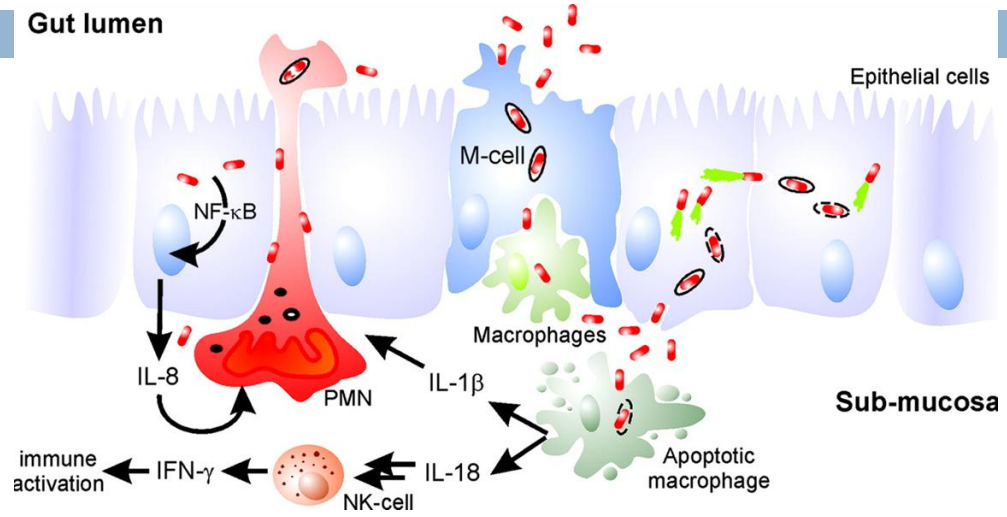
D. *Shigella dysenteriae*

E. *Vibrio cholerae*

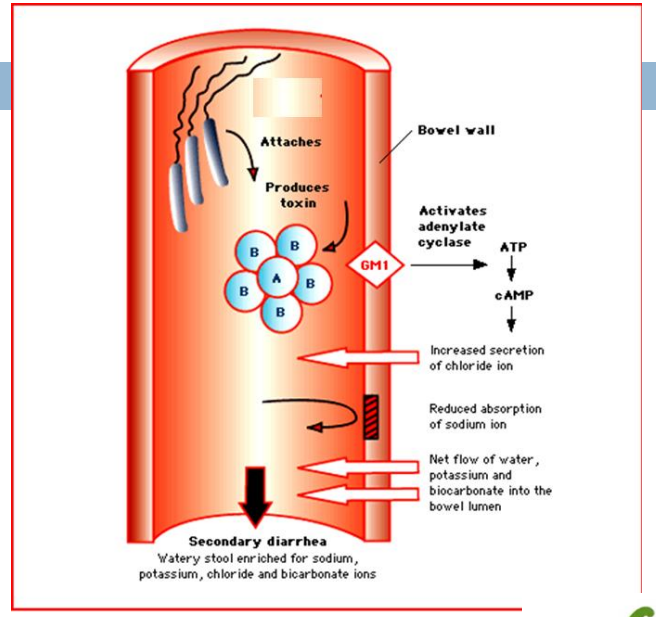
Answer: E.

Which of the following is the most likely pathogenesis?

A.

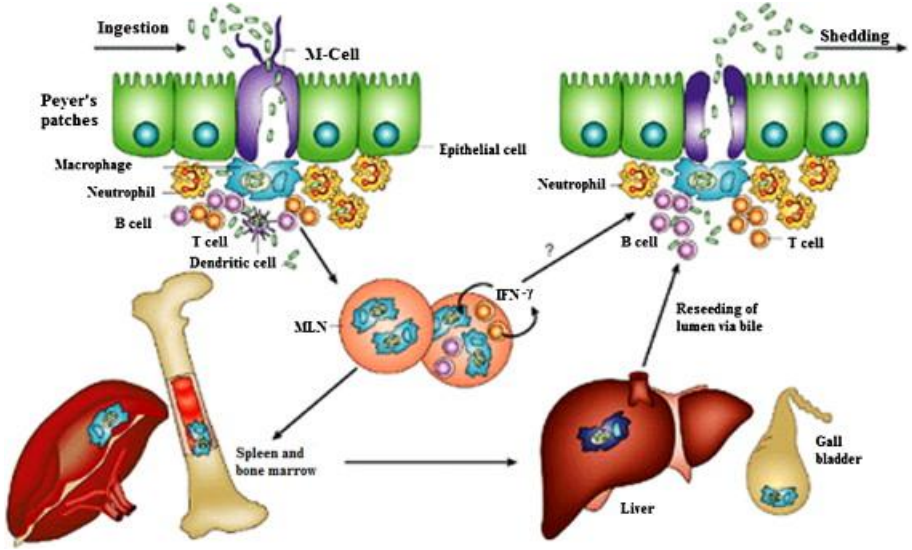


B.

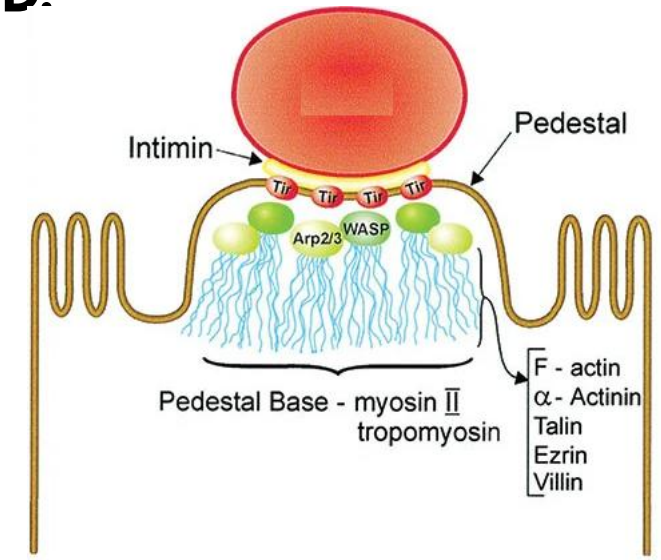


Answer: B.

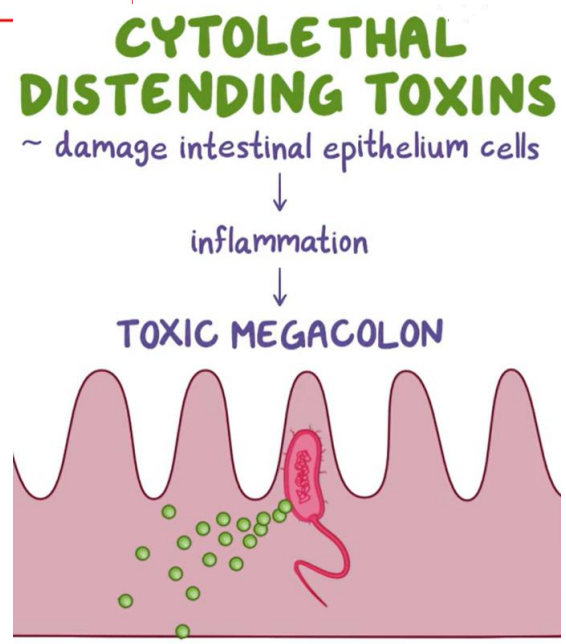
C.



D.



E.



What is the immediate management priority for this patient?

A. Administer antimotility drugs
B. Start antibiotics before fluid resuscitation
C. Administer oral rehydration solution and observe only
D. Begin immediate intravenous fluid resuscitation
E. Administer a proton pump inhibitor

Answers: D

CHOLERA

~ CONTAGIOUS, SEVERE GASTROENTERITIS

~ CAUSED by GRAM-NEGATIVE **VIBRIO CHOLERAE** BACTERIA

TRANSMISSION:

FECAL → ORAL

↓

CONTAMINATED WATER & SHELLFISH





SYMPTOMS:



* VOLUMINOUS "RICE-WATER" DIARRHEA

↓

DEHYDRATION & ELECTROLYTE IMBALANCES

↓

FATAL if NOT TREATED with REHYDRATION SALTS

* SOME STRAINS

↓

CHOLERA ENTEROTOXIN at the SURFACE of INTESTINAL EPITHELIAL CELLS → OVER-ACTIVATION of cAMP by PERMANENTLY ACTIVATING Gs ALPHA



BEFORE **DIAGNOSIS** can be made →



→ growing V. CHOLERAE on THIOSULFATE-CITRATE-BILE SALTS-SUCROSE AGAR

↓

TREATMENT:

~ REPLACE LOST WATER & ELECTROLYTES

└ REHYDRATION SALTS





* MILD-MODERATE DIARRHEA → may RESOLVE in 3-7 days

* EXTREME CASES → ANTIBIOTICS



~ stool culture → which antibiotics are most effective

tetracyclines, ciprofloxacin, ofloxacin, furazolidone, or trimethoprim-sulfamethoxazole

OSMOSIS.org
2022 Edition

Case 4

- A 30-year-old woman develops abrupt nausea, repeated vomiting, and abdominal cramps 3 hours after eating fried rice that had been left at room temperature and reheated.



Which of the following is the most likely causative agent?

A. Shiga toxin of STEC

B. Cholera toxin

C. Enterotoxigenic *E. coli*

D. Preformed heat-stable enterotoxin of *Bacillus cereus*

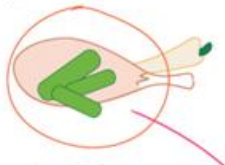
E. Heat-labile enterotoxin of *Staphylococcus aureus*

Answer: D

Which of the following is the most likely pathogenesis?

A.

Heat-labile enterotoxin
Intestinal epithelial cells –> Increases [cAMP]



Meat, vegetables

Bacteria multiplies in GI tract.

Nausea, abdominal cramps, watery diarrhea

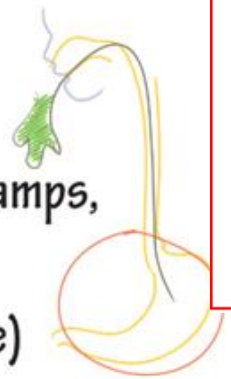
Longer:
Incubation > 6 hrs;
Duration > 24 hrs

B.

Heat-stable toxin
Spores survive initial cooking;
Germinate if not refrigerated.
Reheating does NOT kill toxin.



Rice



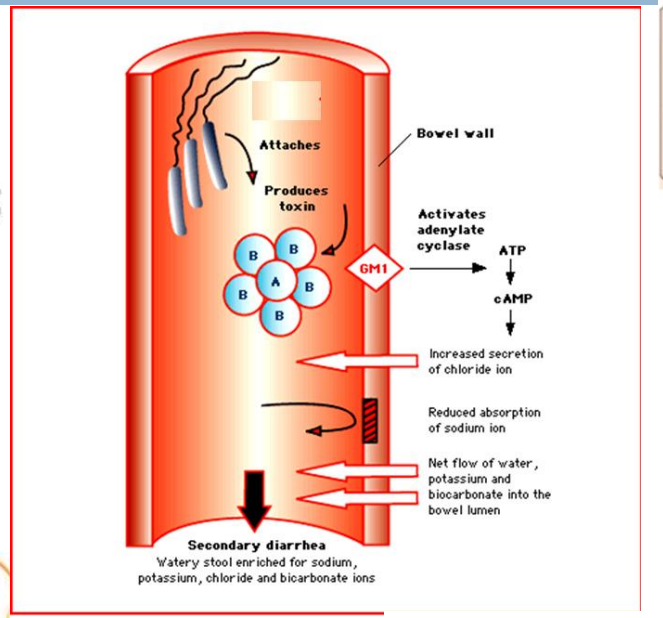
Nausea, abdominal cramps, vomiting.



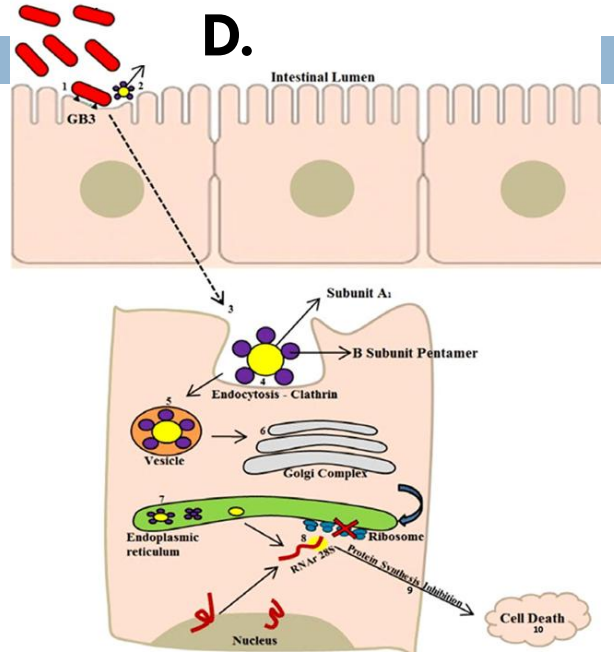
Complication:
Liver failure (rare)

Quick:
Incubation < 6 hrs;
Duration < 24 hrs

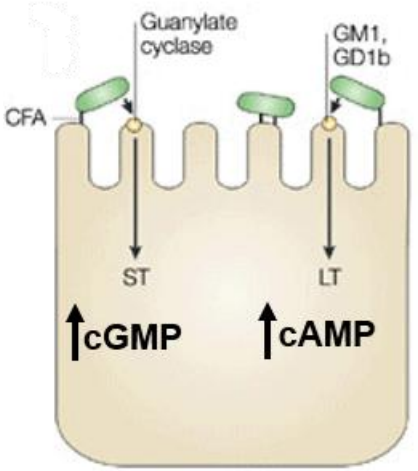
C.



D.



E.



Answer: B.

Incubation Period/ Organisms	Symptoms	Common Food Sources
1 - 6 h **Preformed toxin**		
Staphylococcus aureus heat-stable enterotoxin	Nausea, vomiting, diarrhea	Ham, poultry, potato or egg salad, mayonnaise, cream pastries
Bacillus cereus heat-stable enterotoxin, short-incubation or emetic form	Nausea, vomiting, diarrhea	Fried rice
8 - 16 h		
Clostridium perfringens heat-labile enterotoxin	Abdominal cramps, diarrhea (vomiting rare)	Beef, poultry, legumes, gravies
B. cereus heat-labile enterotoxin, long-incubation or diarrheal form	Abdominal cramps, diarrhea (vomiting rare)	Meats, vegetables, dried beans, cereals
> 16 h		
Vibrio cholerae	Watery diarrhea	Shellfish
Enterotoxigenic *Escherichia coli*	Watery diarrhea	Salads, cheese, meats, water
Enterohemorrhagic *E. coli*	Bloody diarrhea	Ground beef, roast beef, salami, raw milk, raw vegetables, apple juice
Salmonella spp.	Inflammatory diarrhea	Beef, poultry, eggs, dairy products
Campylobacter jejuni	Inflammatory diarrhea	Poultry, raw milk
Shigella spp.	Dysentery	Potato or egg salad, lettuce, raw vegetables
Vibrio parahaemolyticus	Dysentery	Mollusks, crustaceans

Bacillus cereus food poisoning: heat-stable vs heat-labile toxins

EMETIC TYPE

Heat-stable toxin (cereulide)

1. Cooked rice left at room temperature
2. Spores survive cooking
   - Reheating does NOT destroy cereulide
3. Bacteria grow in food and produce preformed toxin
   Cereulide produced in food
   - Incubation: 1 – 6 h
4. Ingestion of contaminated food
5. Cereulide irritates intestinal mucosa → stimulates vagal afferents → activates vomiting center in the brain
   - Rare severe complication: liver failure
6. Nausea and vomiting
   - Main symptom: VOMITING

VS

DIARRHEAL TYPE

Heat-labile enterotoxins (Hbl, Nhe, CytK)

1. Contaminated food (meat, vegetables, sauces, mixed foods)
2. Ingestion of vegetative cells or spores
   - Toxin is heat-labile
3. Bacteria reach small intestine and germinate/multiply
   - Inactivated by heating
4. Enterotoxins produced in the intestine
   - Incubation: 8 – 16 h
5. Toxins bind to intestinal epithelial cells → form pores → fluid secretion into intestinal lumen
6. Watery diarrhea and abdominal cramps
   - Main symptom: DIARRHEA

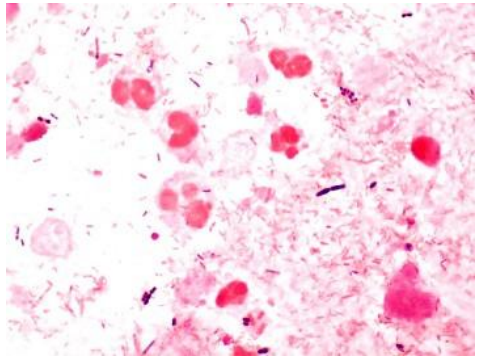
Comparison	Where toxin is made		Heat property		Typical food		Main symptom	
	In food (before ingestion)	In intestine (after ingestion)	Heat-stable (cereulide)	Heat-labile (Hbl, Nhe, CytK)	Often rice / starchy foods	Various foods	Vomiting	Watery diarrhea

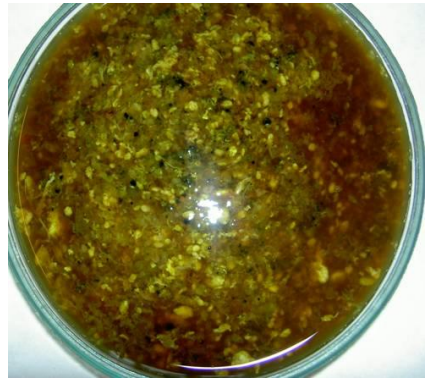
Major heat-labile enterotoxins of *B. cereus*: Hbl = **hemolysin BL,** Nhe = **non-hemolytic enterotoxin,** and CytK = **cytotoxin K**

AI-assisted figure preparation: ChatGPT, OpenAI, was used to generate and refine the visual flow-chart layout. The scientific content was reviewed and edited by the instructor.

Case 5

- A 65-year-old man develops fever, abdominal cramps, nausea, and diarrhea with mucus 24 hours after eating undercooked eggs. Stool microscopy shows 50–100 neutrophils/HPF and no erythrocytes. He is alert and hemodynamically stable, has no history of vascular grafts or immunosuppression, and has not recently taken antibiotics.





Which of the following is the most likely causative agent?

A. Enterotoxigenic *E. coli*

B. *Clostridioides difficile*

C. *Vibrio cholerae*

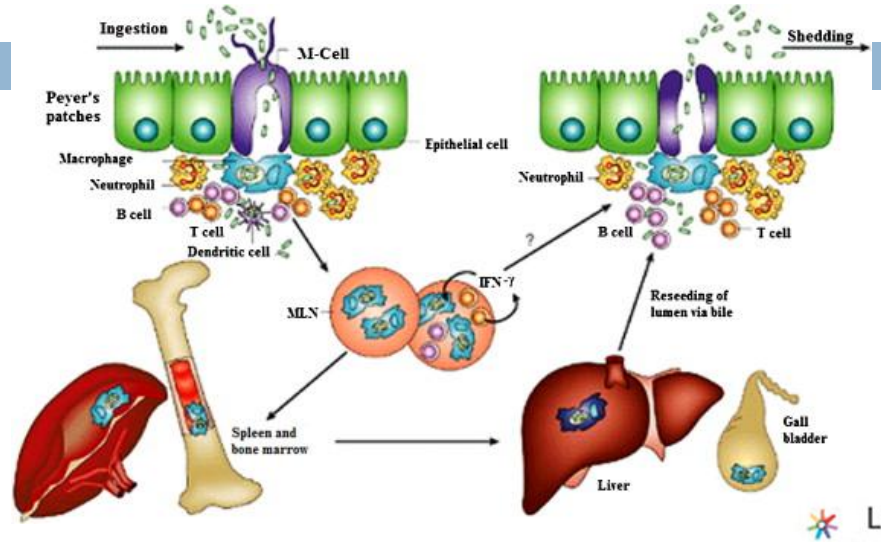
D. *Salmonella* Typhi

E. Non-typhoidal *Salmonella*

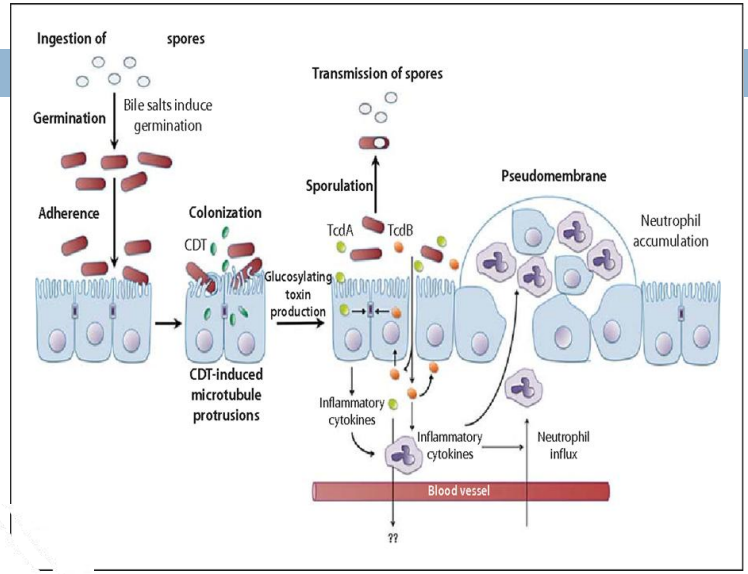
Answer: E.

Which of the following is the most likely pathogenesis?

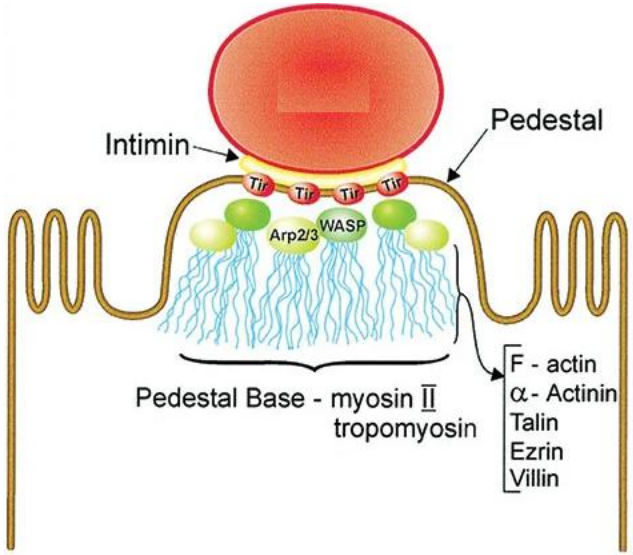
A.



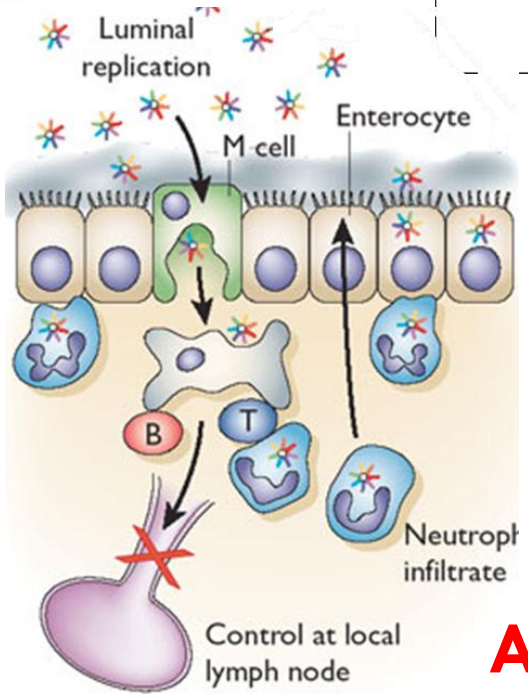
B.



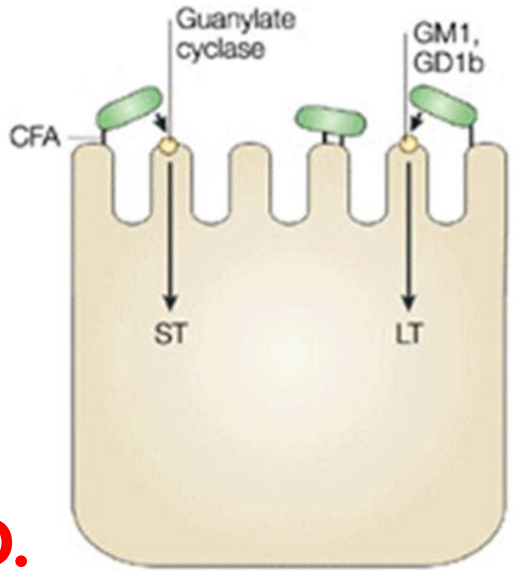
C.



D.

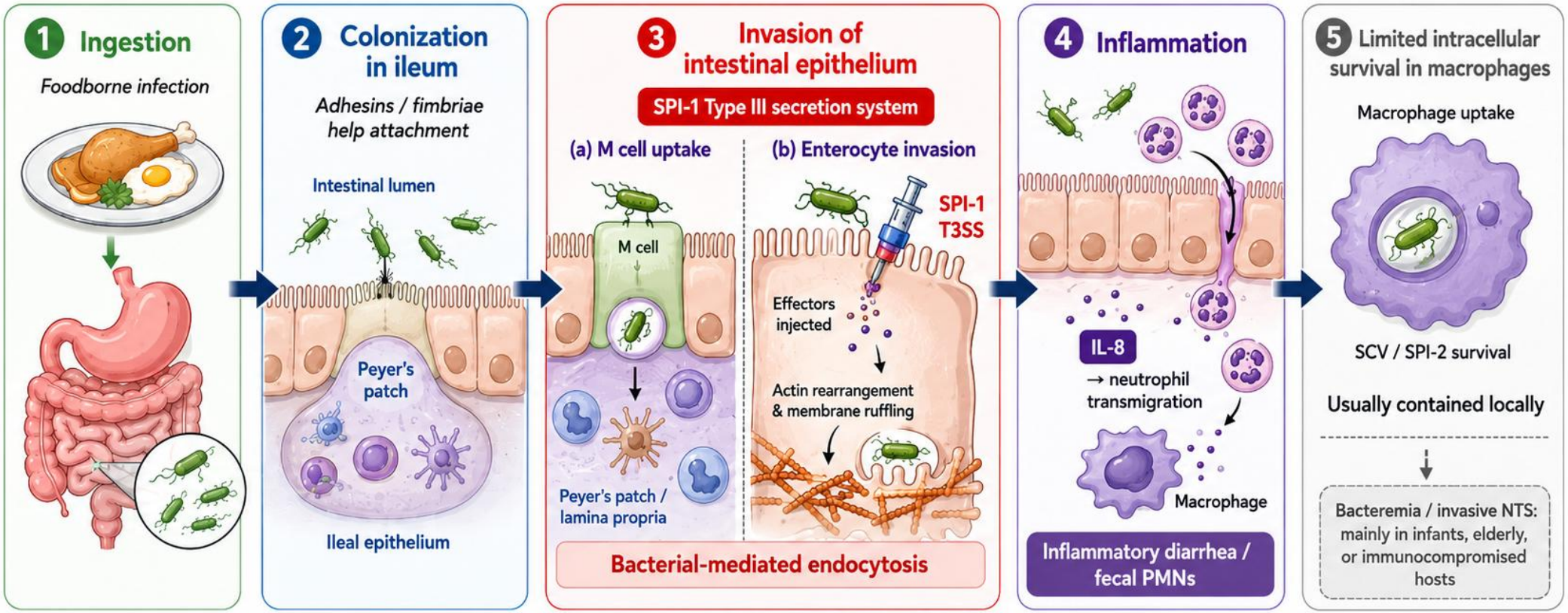


E.



Answer: D.

Mechanism & Pathogenesis of Non-typhoidal *Salmonella*





Adapted from standard medical microbiology and review sources.

AI-assisted figure preparation: ChatGPT, OpenAI, was used to generate and refine the visual flow-chart layout. The scientific content was reviewed and edited by the instructor.

What is the appropriate treatment?

A. Activated charcoal

B. Antibiotic administration

C. Antidiarrheal drug

D. Fluid and electrolyte replacement

E. Proton pump inhibitor

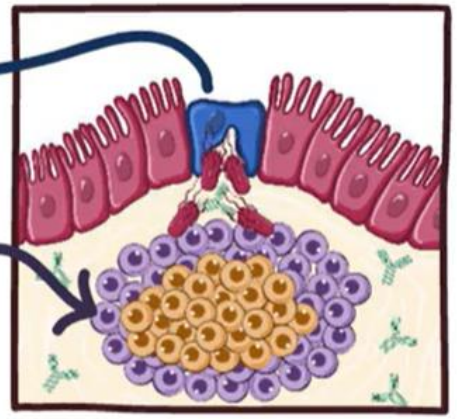
Answer: D.

SALMONELLA



* ROD SHAPED
* GRAM NEGATIVE
* ENTEROBACTERIACEAE FAMILY

* INFECT M-CELLS
  ↓
  PEYER'S PATCHES



NON-TYPHOIDAL SEROTYPES

* STRONG PRO-INFLAMMATORY RESPONSE
  ↳ INFLAMMATION of SMALL INTESTINE and COLON
* INFECTIONS are FOODBORNE or from INTERACTIONS with ANIMAL RESERVOIRS

SYMPTOMS of "FOOD POISONING"

* ABDOMINAL CRAMPS
* VOMITING
* HEADACHES
* FEVER
* DIARRHEA





DIAGNOSIS

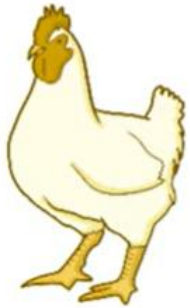
* STOOL CULTURE

TREATMENT

Self-limiting; ATB: can prolong carrier state or cause relapse

* UNCOMPLICATED - NON INVASIVE
  - FLUID AND ELECTROLYTE REPLENISHMENT
* COMPLICATED - INVASIVE
  - ANTIBIOTIC THERAPY







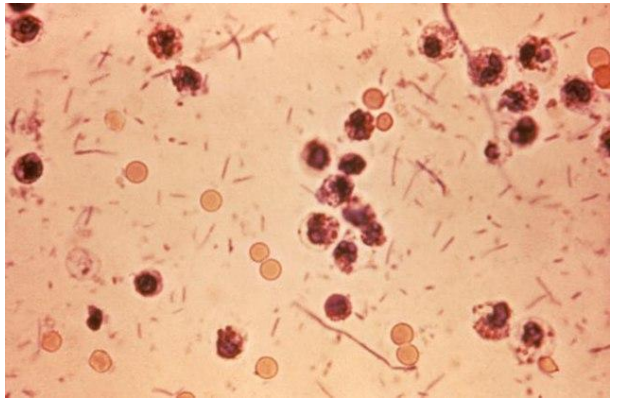
OSMOSIS.org
2022 Edition

Case 6



- A 15-year-old girl has 3 days of bloody diarrhea and abdominal pain after eating a hamburger. She receives an antibiotic. One week after diarrhea began, she develops pallor and reduced urine output. Tests show microangiopathic hemolytic anemia, thrombocytopenia, and elevated creatinine.





What is the most likely diagnosis?

A. Acute glomerulonephritis

B. Post-infectious nephritis

C. Hemolytic Uremic Syndrome (HUS)

D. Idiopathic thrombocytopenic purpura (ITP)

E. Thrombotic thrombocytopenic purpura (TTP)

Answer: C.

What is the most likely causative agent?

A. *Streptococcus pyogenes*

B. Shiga toxin-producing *Escherichia coli* (STEC)

C. *Shigella flexneri*

D. *Clostridioides difficile*

E. *Campylobacter jejuni*

Answer: B.

What is the most likely pathophysiological mechanism?

A. Immune-complex deposition in glomeruli

B. Shiga toxin-mediated renal microvascular endothelial injury

C. Direct bacterial invasion of the kidney

D. Autoantibodies against red blood cells

E. Viral infection of renal epithelial cells

Answer: B.

What is the mechanism that best explains the worsening of his symptoms?

A. Selection of an antibiotic-resistant STEC strain

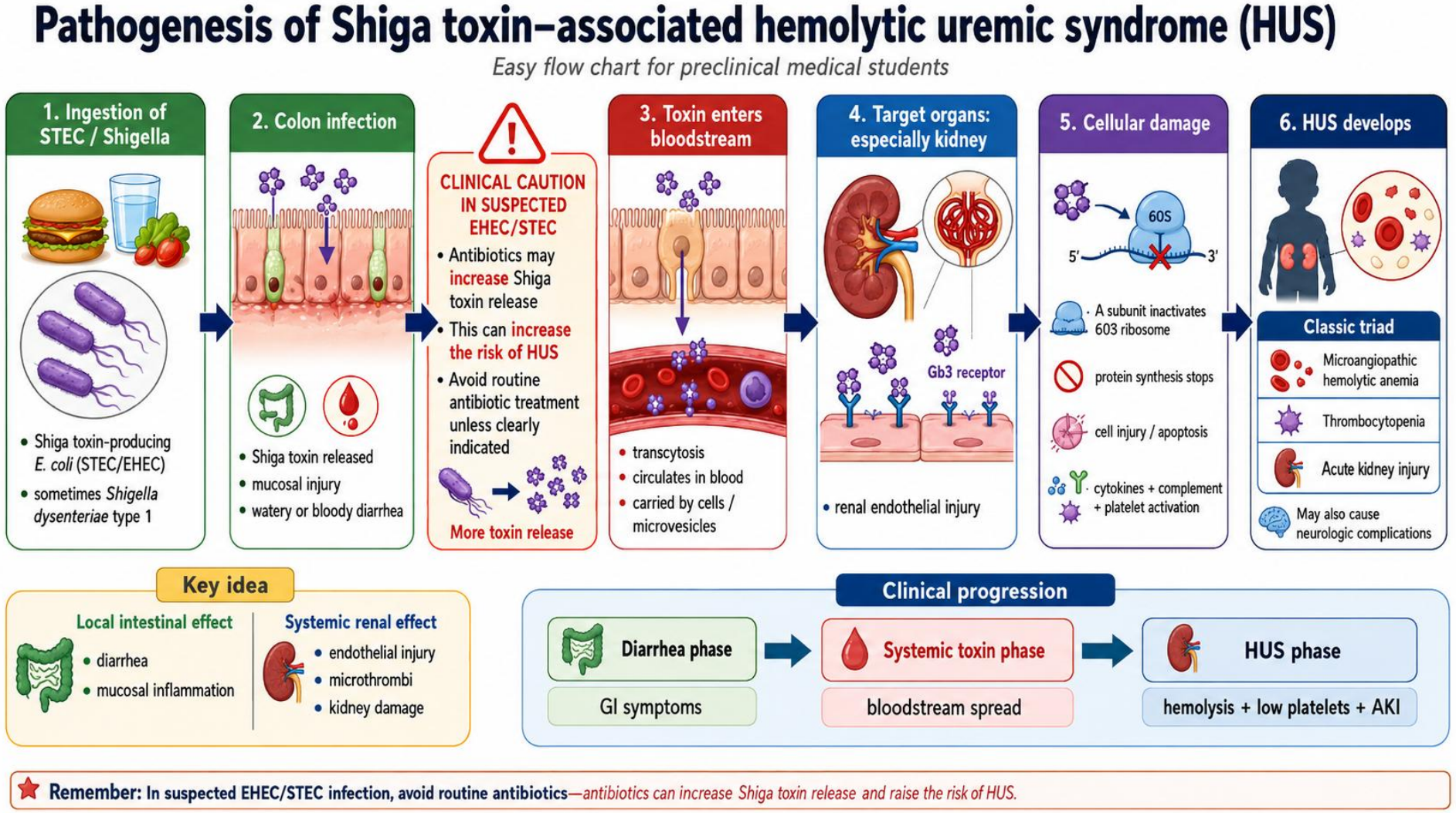
B. Direct antibiotic destruction of red blood cells

C. Direct antibiotic toxicity to glomerular cells

D. Increased expression or release of Shiga toxin

E. Immune-complex deposition in the kidney

Answer: D.

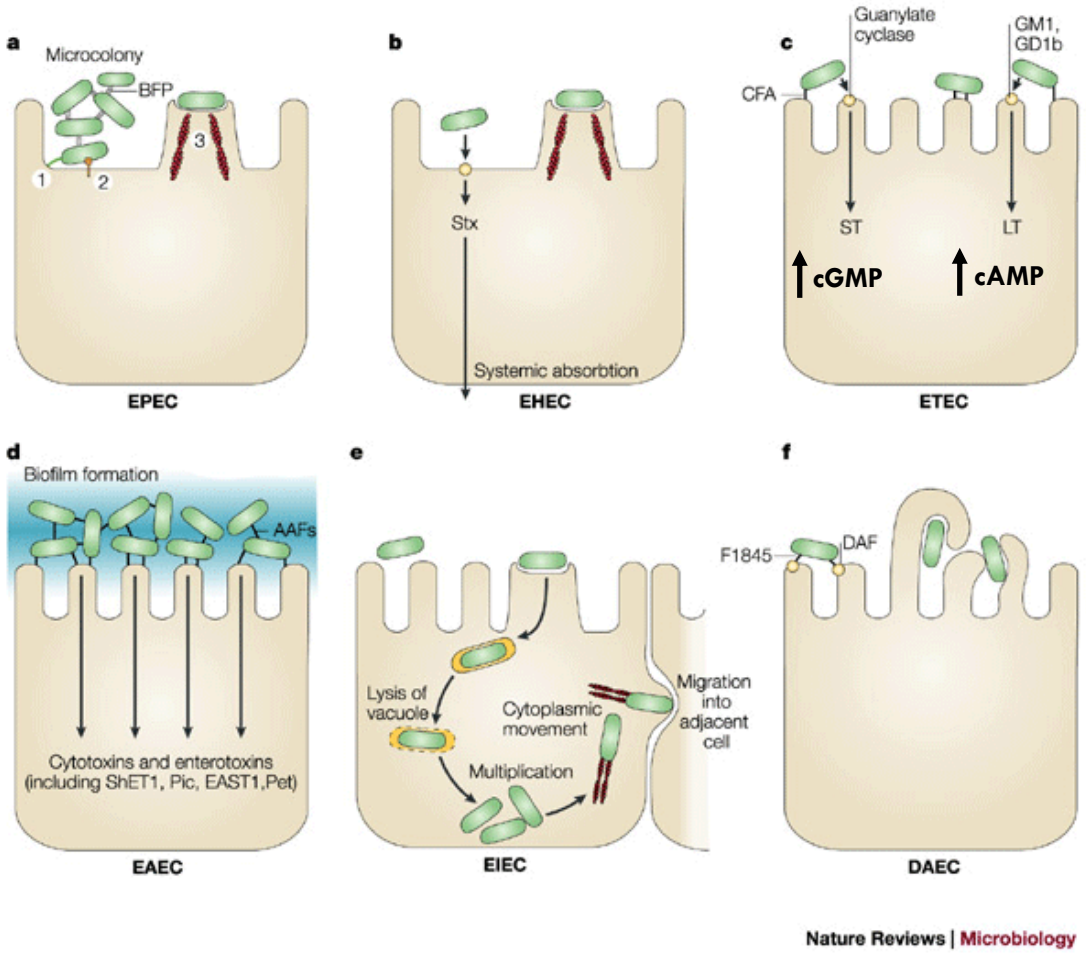


- Shiga toxin-producing *E. coli* (STEC), including O157 and non-O157 strains
- *Shigella dysenteriae* type 1
- EHEC >>> *Shigella*

- Circulating Shiga toxin injures renal microvascular endothelium.
- Antibiotics are generally avoided when STEC is suspected because they may increase HUS risk.
- Monitor hemoglobin, platelets, and kidney function; provide supportive care.

AI-assisted figure preparation: ChatGPT, OpenAI, was used to generate and refine the visual flow-chart layout. The scientific content was reviewed and edited by the instructor.

Diarrheagenic *E. coli*: pathotypes



- **Watery diarrhea**

- EPEC: enteropathogenic
- ETEC: enterotoxigenic
- EAEC: enteroaggregative
- DAEC: diffusely adherent

- **Inflammatory diarrhea**

- EHEC: enterohemorrhagic [aka Shiga toxin-producing *E. coli* (STEC), verocytotoxin-producing *E. coli* (VTEC)]
- EIEC: enteroinvasive

Nature Reviews Microbiology volume 2, pages123–140 (2004)

Complications of bacterial enteric infection

Complication	Important Bacterial Agents	Clinical Considerations
Dehydration	*Vibrio cholerae*, any bacterial enteropathogen	Most important complication of all forms of acute watery diarrhea; should prompt aggressive fluid and electrolyte replacement, usually in hospital
Bacteremia	Salmonella, *Campylobacter fetus*	Organisms that deeply penetrate the intestinal mucosa are prone to cause bacteremia; certain high-risk conditions predispose to systemic salmonella infection
Hemolytic–uremic syndrome	Shiga toxin–producing *Escherichia coli*	Shiga toxin is absorbed, causing injury to endothelial cells of the glomerular capillaries with intravascular coagulation
Guillain–Barré syndrome	*Campylobacter jejuni*	Most cases occur as a result of molecular mimicry, with antibodies directed to campylobacter lipooligosaccharides and peripheral-nerve gangliosides; probability of development of Guillain–Barré syndrome within 2 mo after campylobacter infection estimated at <2/10,000 cases[40]

Complications of bacterial enteric infection

Complication	Important Bacterial Agents	Clinical Considerations
Reactive arthritis and iritis	Campylobacter, salmonella, *Shigella flexneri*, yersinia	Occurs in 2.1/100,000 cases of campylobacter infection and 1.4/100,000 cases of salmonella infection; affected persons may be HLA-B27–positive or HLA-B27–negative[41]
Postinfectious irritable bowel syndrome	Inflammatory bacterial pathogens (e.g., campylobacter) are most important, but most bacterial pathogens can produce the syndrome	Enteric bacterial infection with intestinal inflammation in a susceptible host leads to altered intestinal findings and postinfectious irritable bowel syndrome[42-44]; duration is ≥5 yr[45-47]

Summary

Mechanism	Location	Illness	Stool Findings	Examples of Pathogens Involved
Noninflammatory (enterotoxin)	Proximal small bowel	Watery diarrhea	No fecal leukocytes; mild or no increase in fecal lactoferrin	*Vibrio cholerae*, enterotoxigenic *Escherichia coli* (LT and/or ST), entero-aggregative *E. coli*, *Clostridium perfringens*, *Bacillus cereus*, *Staphylococcus aureus*, *Aeromonas hydrophila*, *Plesiomonas shigelloides*, rotavirus, noro-virus, enteric adenoviruses, *Giardia lamblia*, *Cryptosporidium* spp., *Cyclospora* spp., microsporidia
Inflammatory (invasion or cytotoxin)	Colon or distal small bowel	Dysentery or inflammatory diarrhea	Fecal polymorphonuclear leukocytes; substantial increase in fecal lactoferrin	*Shigella* spp., *Salmonella* spp., *Campylobacter jejuni*, enterohemorrhagic *E. coli*, enteroinvasive *E. coli*, *Yersinia enterocolitica*, *Listeria monocytogenes*, *Vibrio parahaemolyticus*, *Clostridium difficile*, *A. hydrophila*, *P. shigelloides*, *Entamoeba histolytica*, *Klebsiella oxytoca*
Penetrating	Distal small bowel	Enteric fever	Fecal mononuclear leukocytes	*Salmonella typhi*, *Y. enterocolitica*

Abbreviations: LT, heat-labile enterotoxin; ST, heat-stable enterotoxin.

Copyright© McGraw-Hill Education. All rights reserved.


Q
A